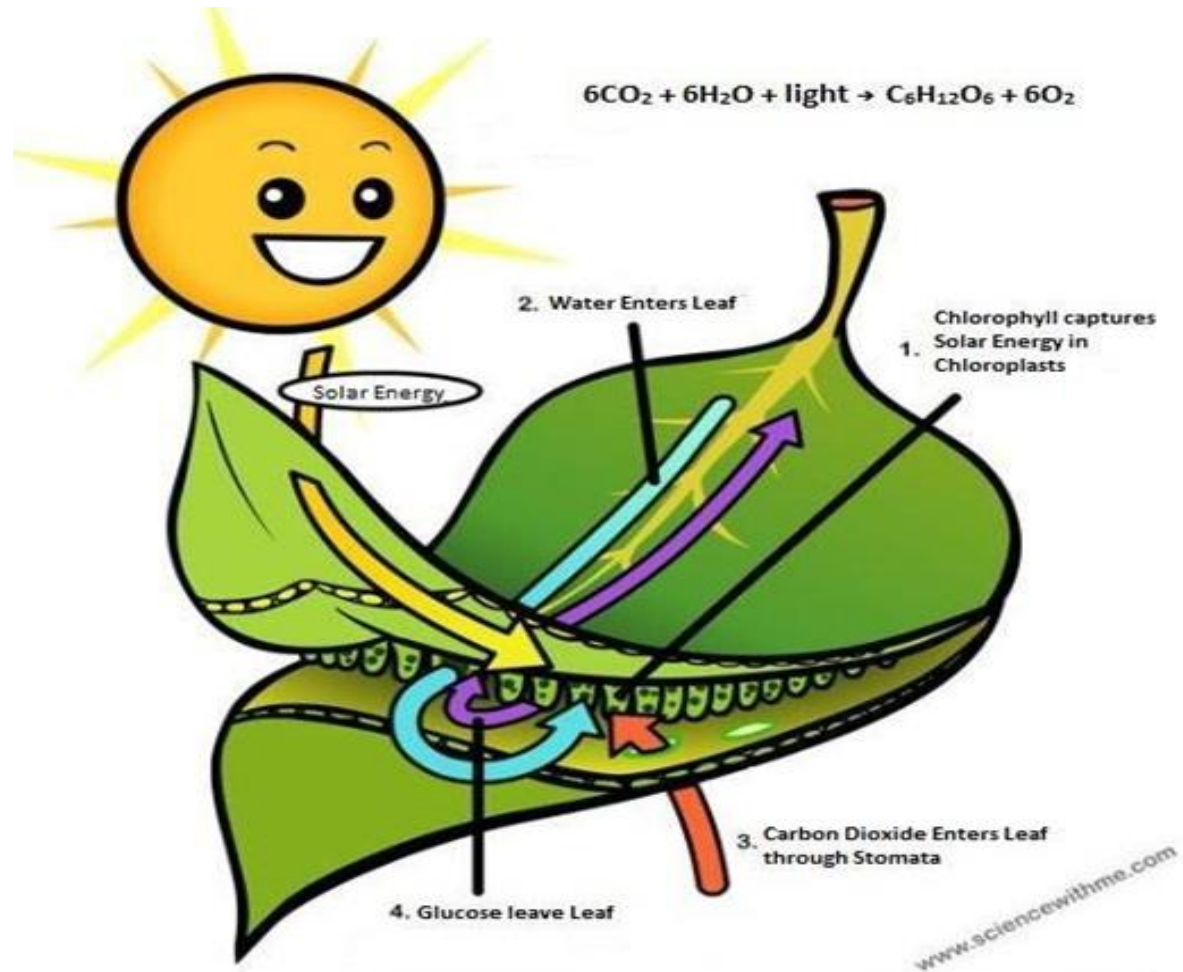


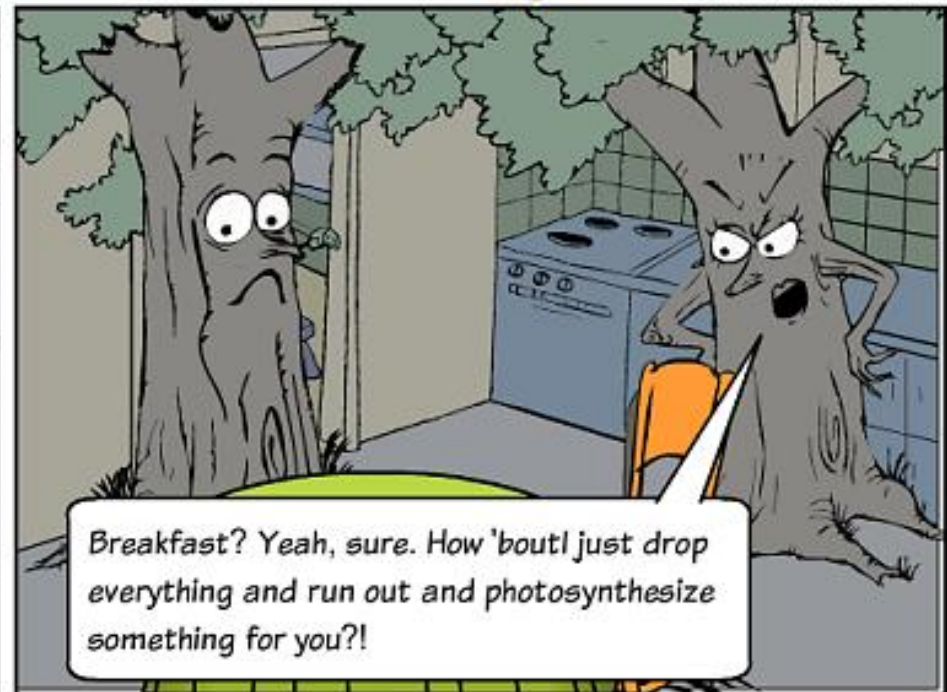
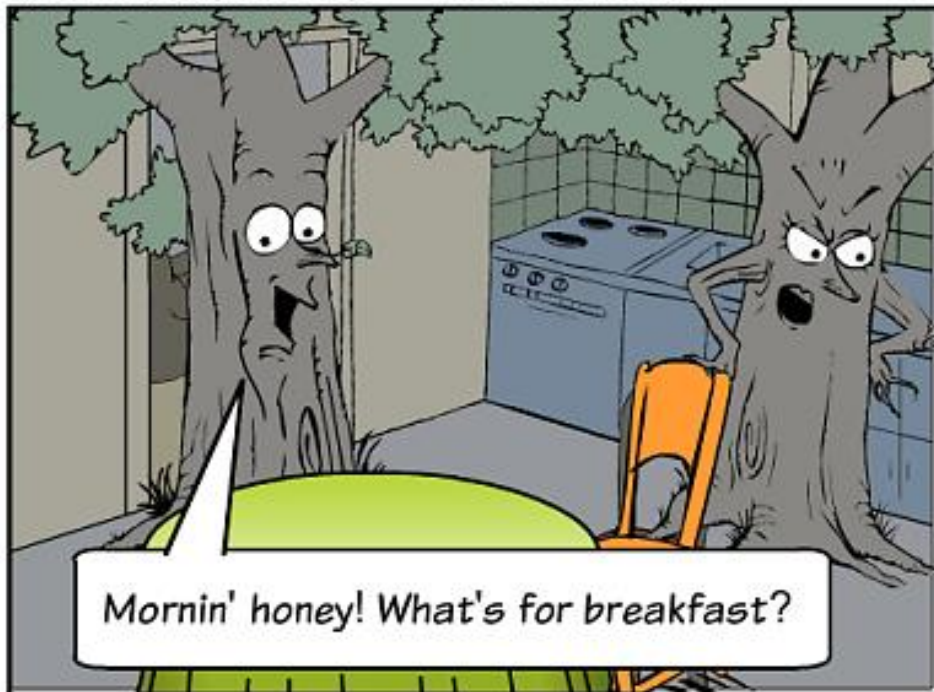
Photosynthesis



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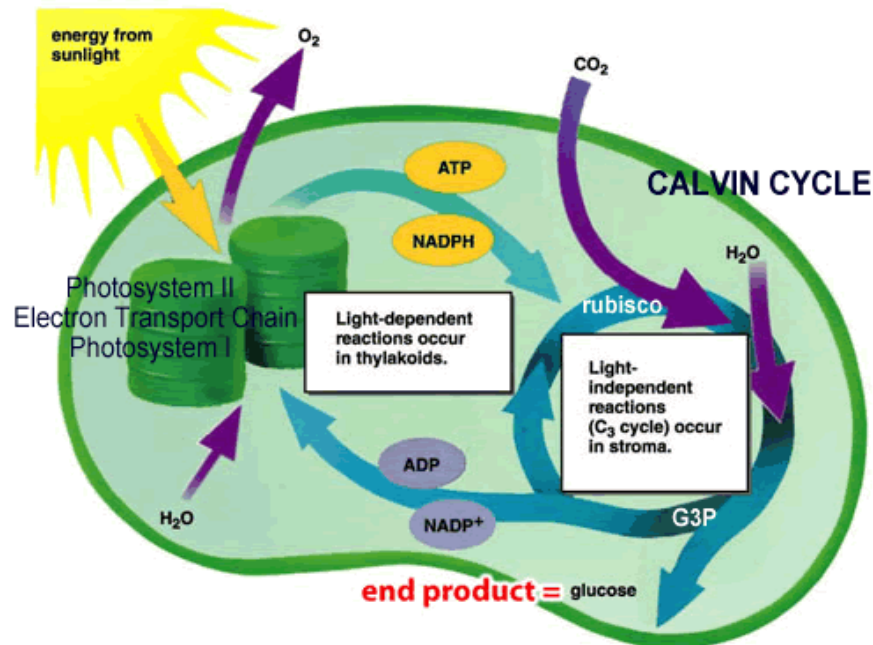
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Photosynthesis

- A process whereby light energy is converted to chemical energy that is stored organic compounds (e.g. sugar)

Light energy



Photosynthesis

- Occurs in plants, algae, certain other protists, and some prokaryotes



(a) Plants

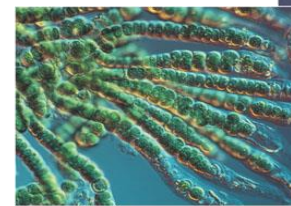


(b) Multicellular algae



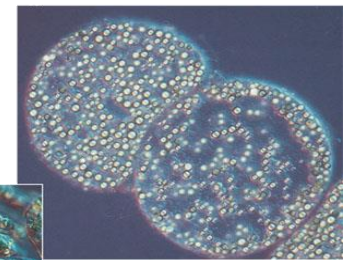
(c) Unicellular protist

10 μm



(d) Cyanobacteria

40 μm

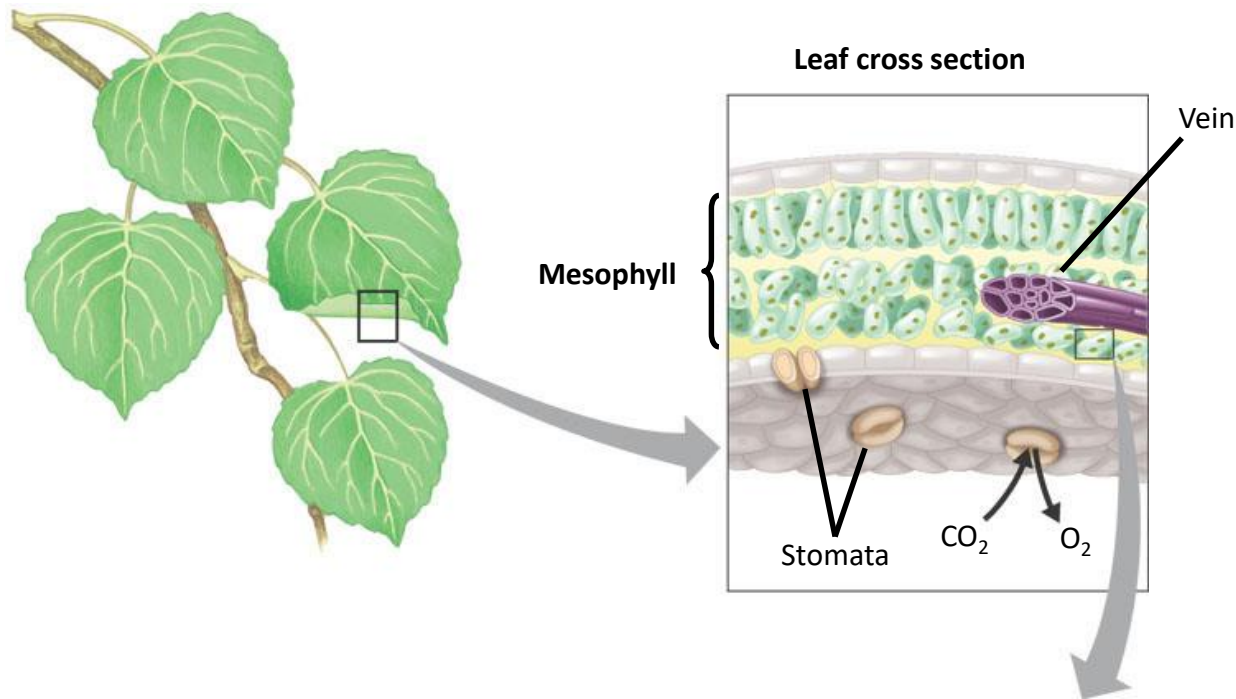


(e) Purple sulfur bacteria

1.5 μm

The Site of Photosynthesis

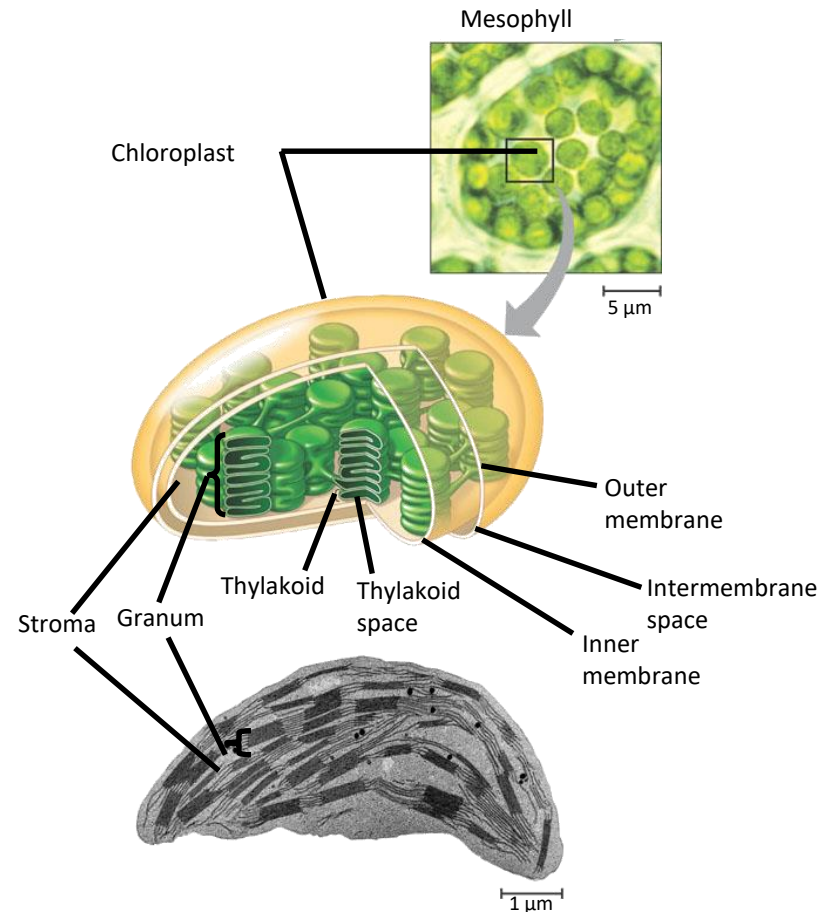
- The leaves of plants
 - Are the major sites of photosynthesis
 - Has mesophyll cells with **chloroplast**



The Site of Photosynthesis:

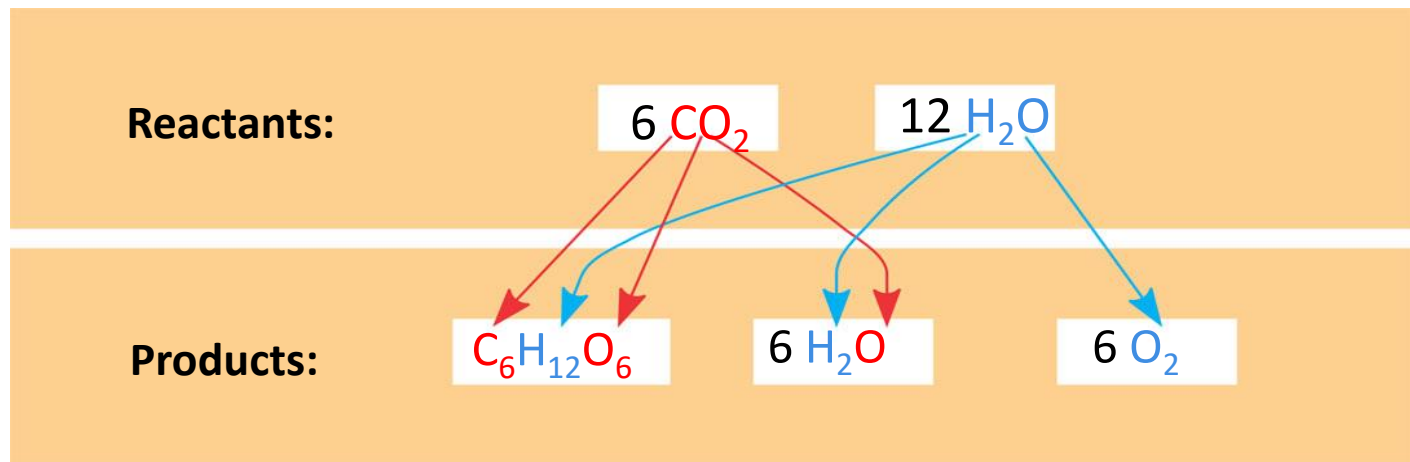
Chloroplasts

- Are the organelles in which photosynthesis occurs
- **Thylakoids:** membranous sacks that contain photosynthetic machinery
- **Granum/grana:** Stack of thylakoids
- **Stroma:** Fluid surrounding the thylakoids



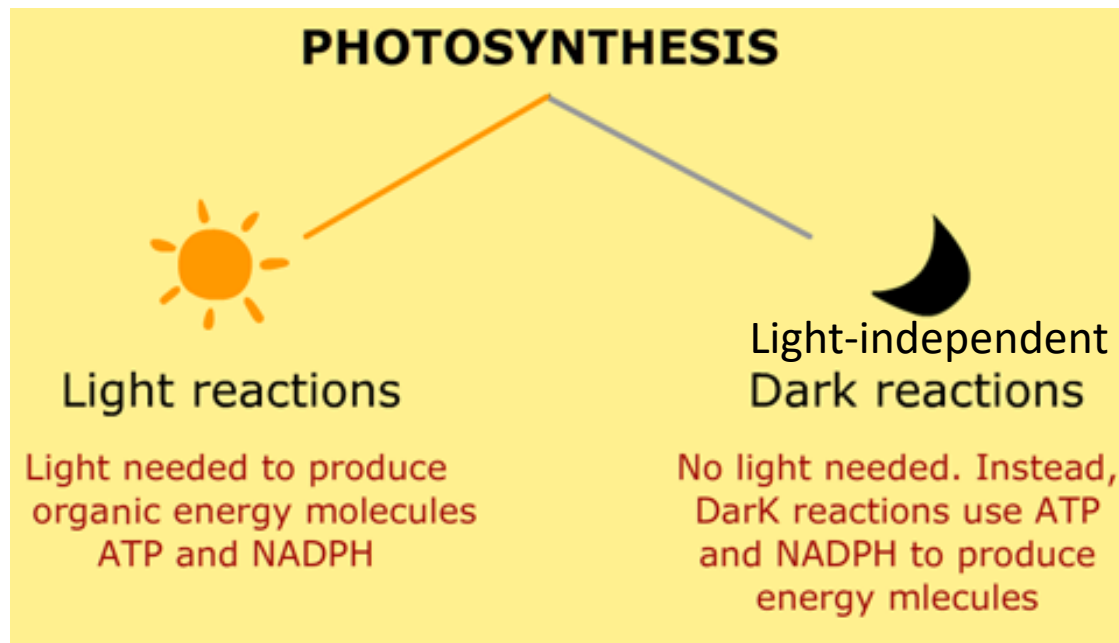
The splitting of water

- Chloroplasts split water into
 - Hydrogen and oxygen, incorporating the electrons of hydrogen into sugar molecules
- Photosynthesis is a redox process
 - Water is oxidized, carbon dioxide is reduced



Photosynthesis: 2 processes

- Light-dependent / light reaction
- Light-independent /dark reaction / Calvin Cycle



Photosynthesis: 2 processes

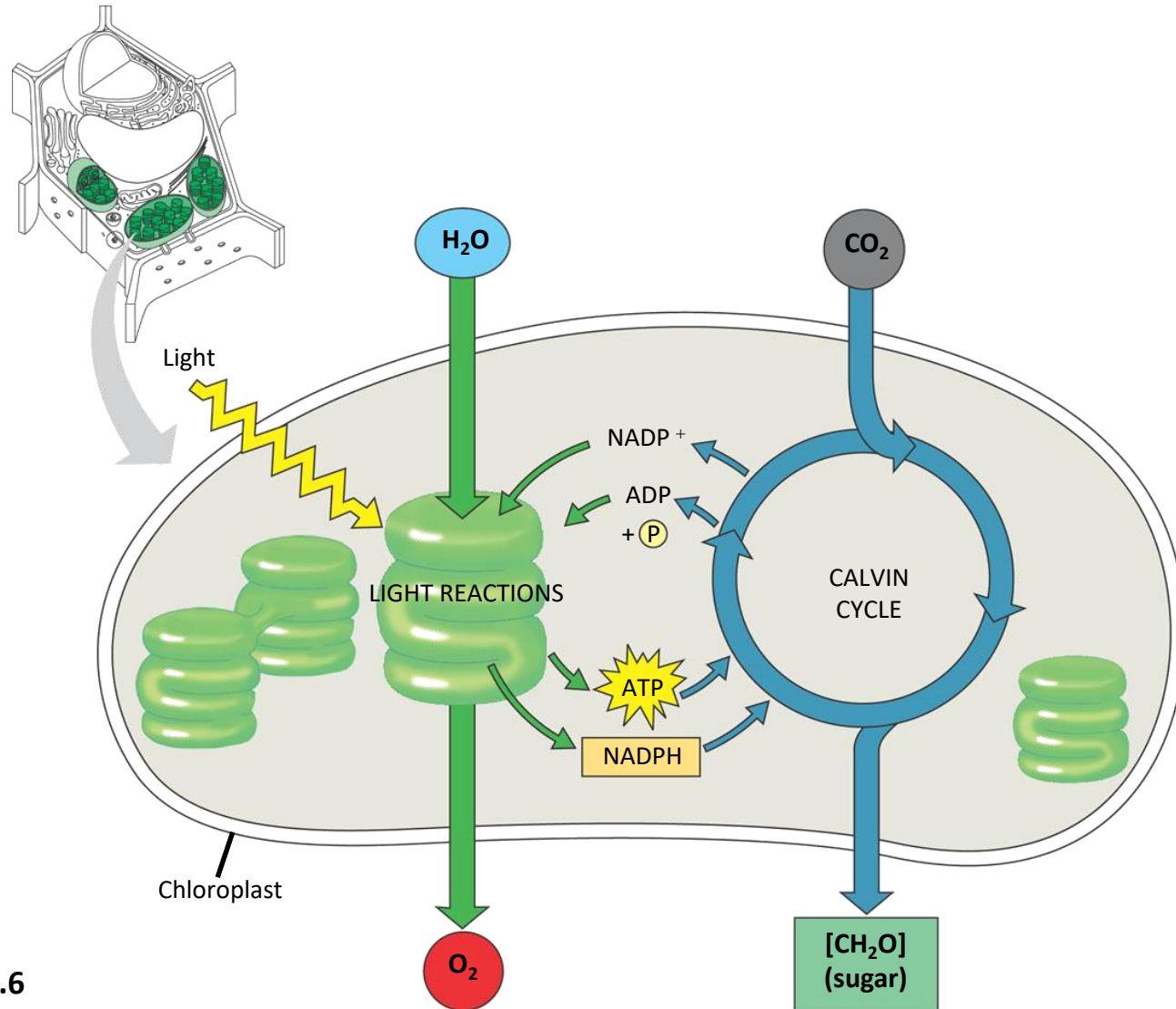


Figure 11.6

Light-dependent reaction

- Occurs in the grana
- Split water, release oxygen, produce ATP and form NADPH

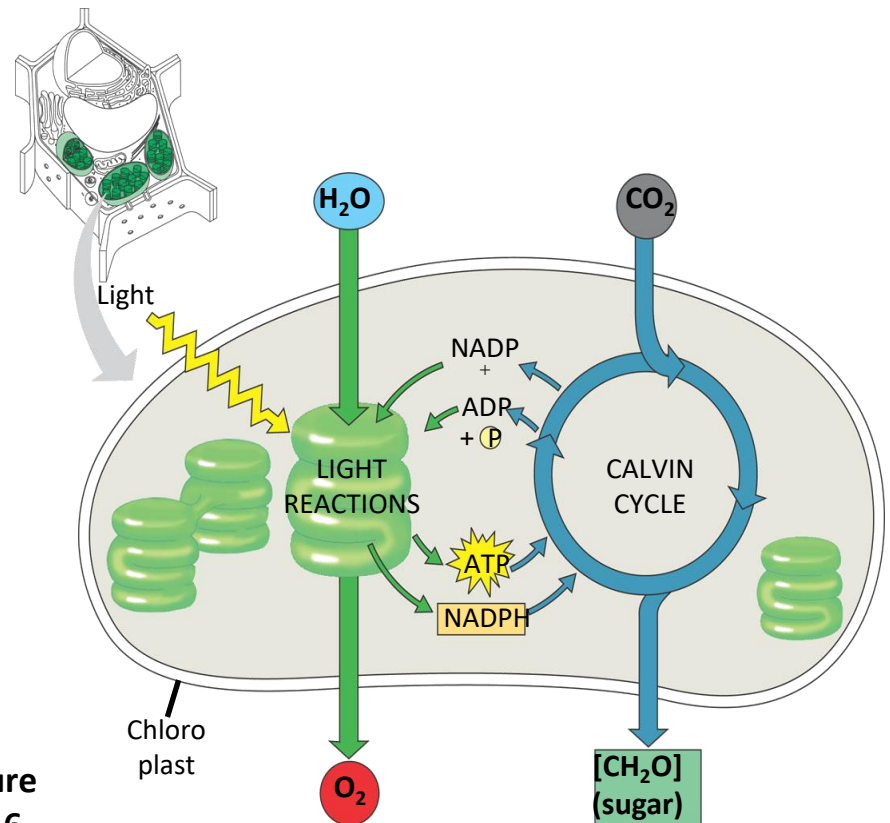


Figure
11.6

Light-independent reaction/Calvin cycle

- Occurs in the stroma
- Forms sugar from carbon dioxide, using ATP for energy and NADPH for reducing power

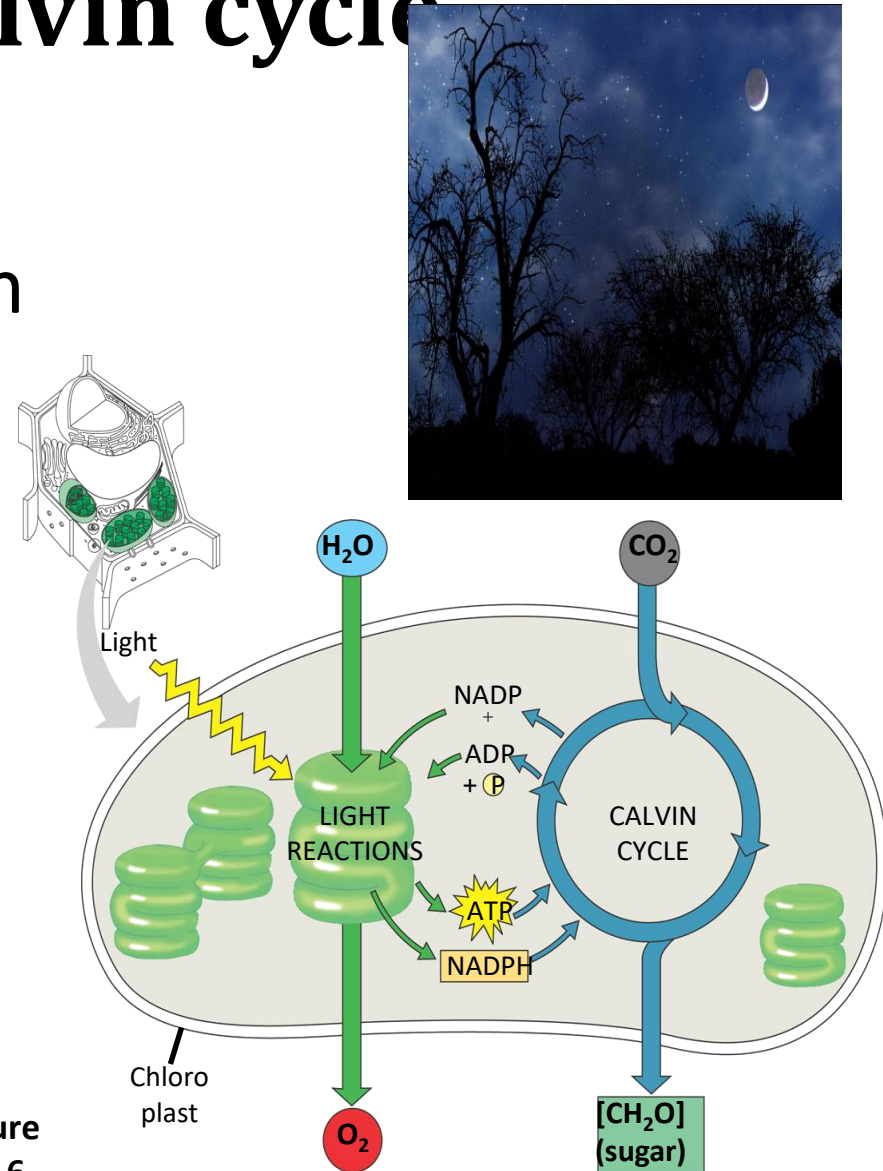


Figure
11.6

The Nature of Sunlight

- Light
 - Is a form of electromagnetic energy, which travels in waves

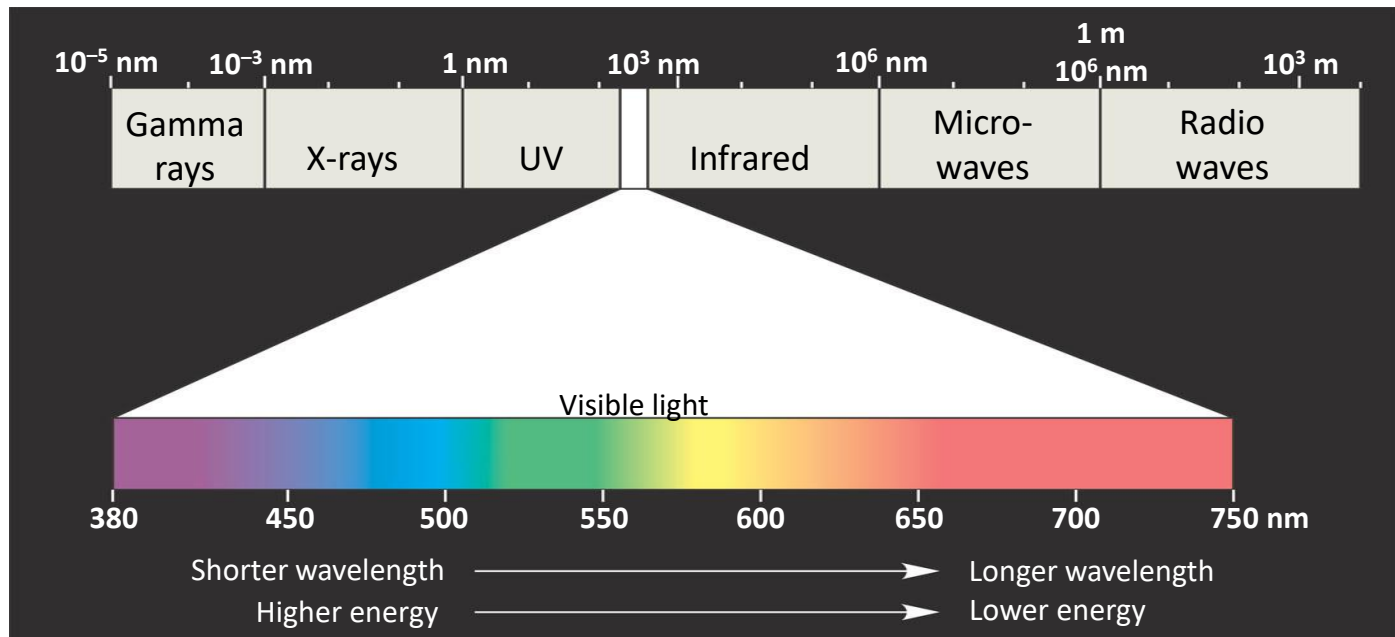
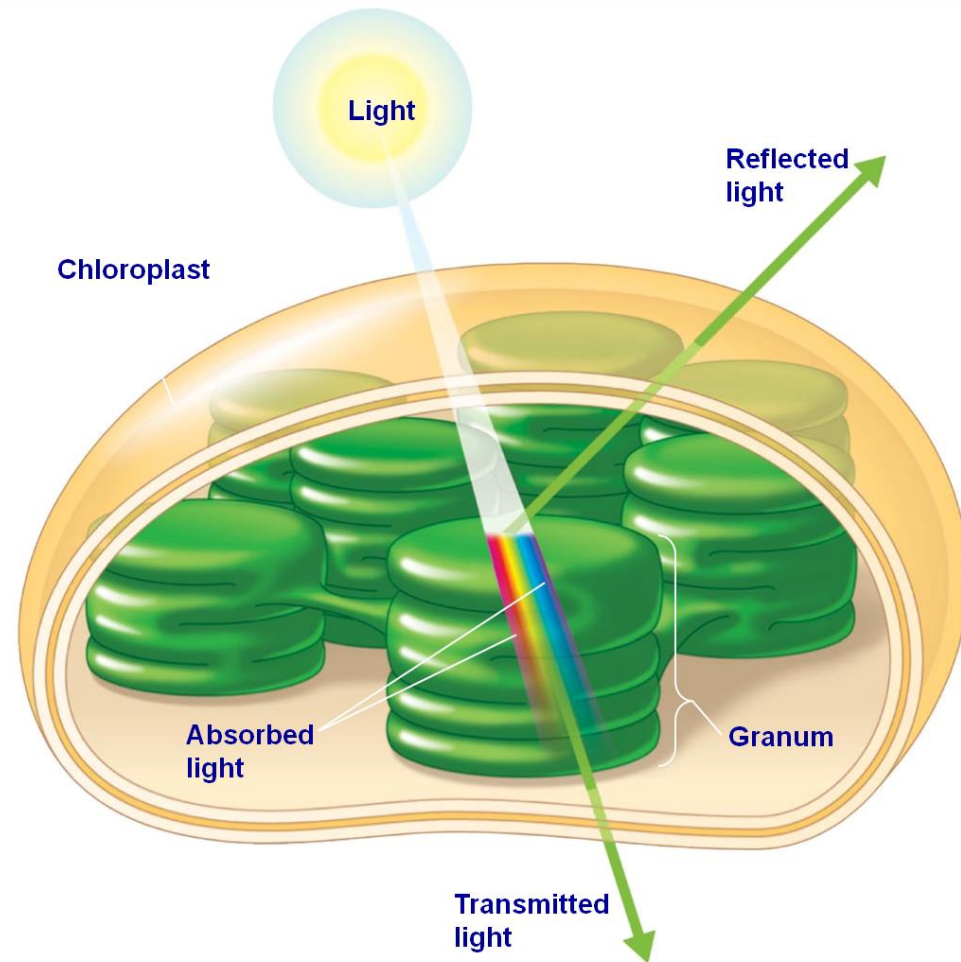


Figure 11.7

Photosynthetic Pigments

- Are substances that absorb visible light
- 2 types:
 - Chlorophylls/primary
 - Carotenoids/accessory



Photosynthetic Pigments

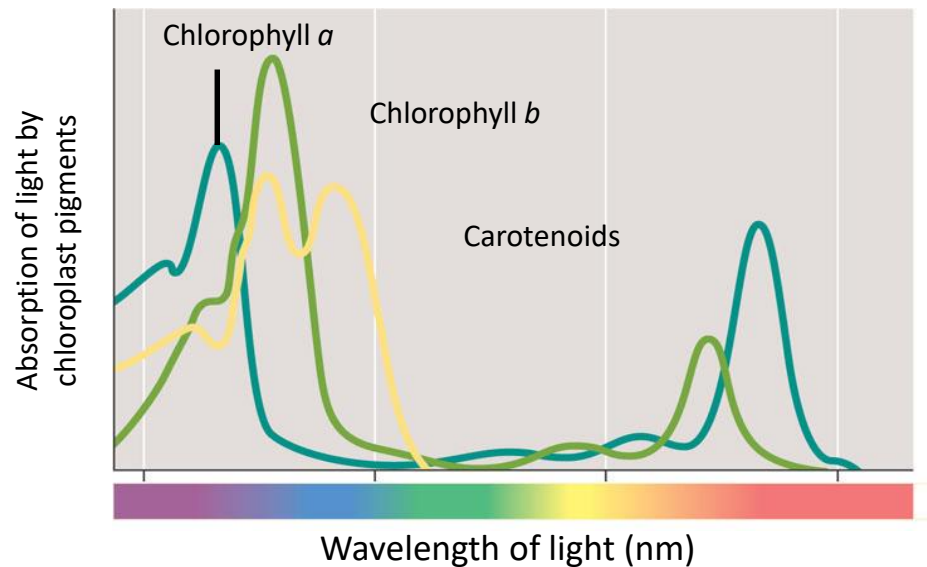
- **Chlorophyll *a* & *b*** – green & absorb all wavelengths of light in the red, blue, violet range
- **Carotenoids** – yellow, orange & red. Absorb light in the blue, green & violet range
 - **Xanthophyll** – type of carotenoids
- **Phycobilins** – reddish, absorb light in the blue & green range

Antenna pigments

- Antenna/accessory pigments : Chlorophyll *b*, carotenoids, phycobilins
- Work in conjunction with chlorophyll *a*
- Capture light in λ different from chlorophyll *a*
- Absorb photon of lights & pass energy along to chlorophyll *a*
- Directly involve in transformation of light energy to sugars

Absorbtion spectra of pigments

RESULTS

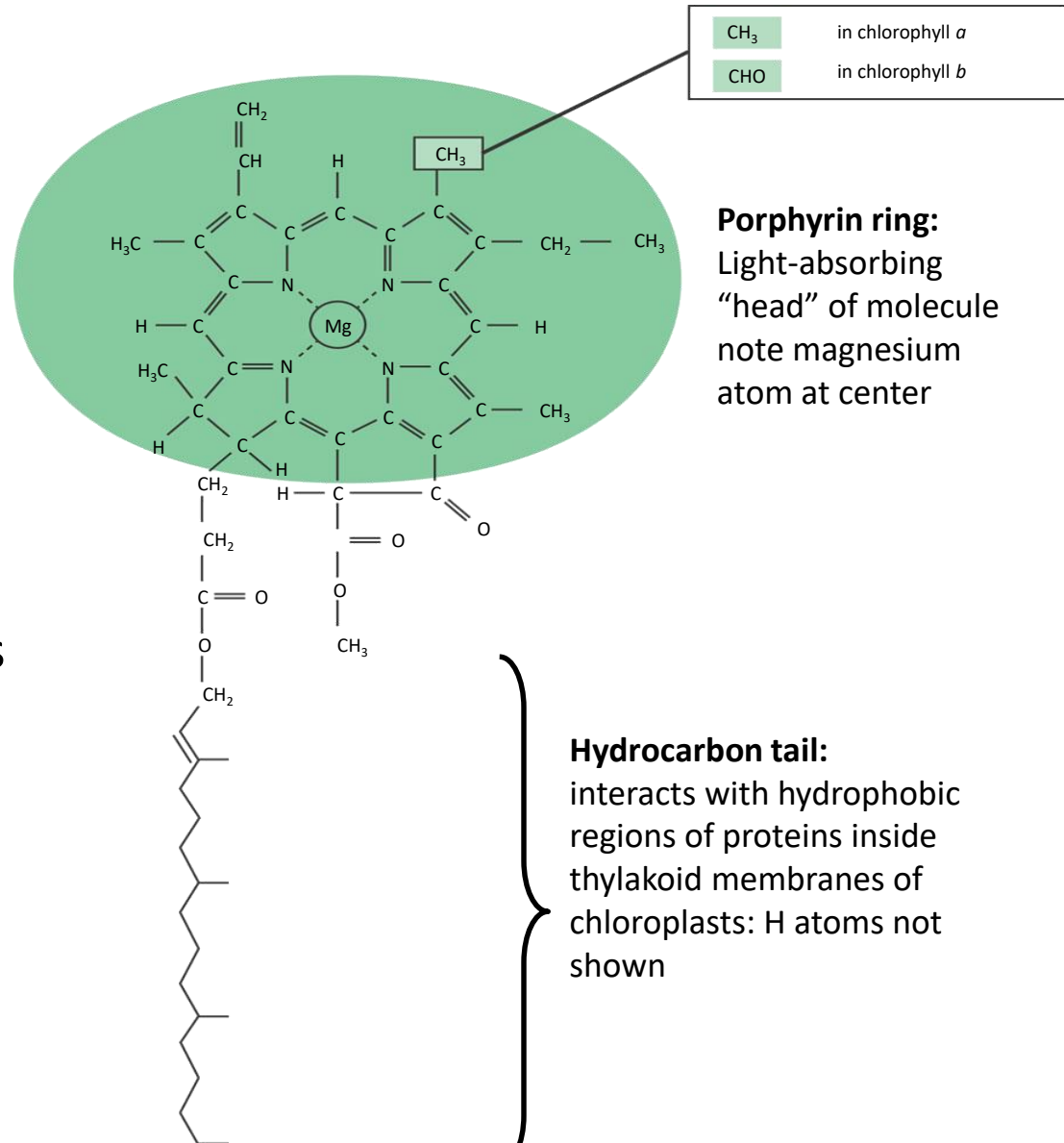


(a) **Absorption spectra.** The three curves show the wavelengths of light best absorbed by three types of chloroplast pigments.

Figure 11.10

Chlorophyll *a*

- Pigments participate directly in light reaction
- Large molecule with a single Mg atom in the head surrounded by alternating double & single bonds
- Head (porphyrin ring) attached to long hydrocarbon tail
- Double bonds as sources of electron flow via ETC during photosynthesis



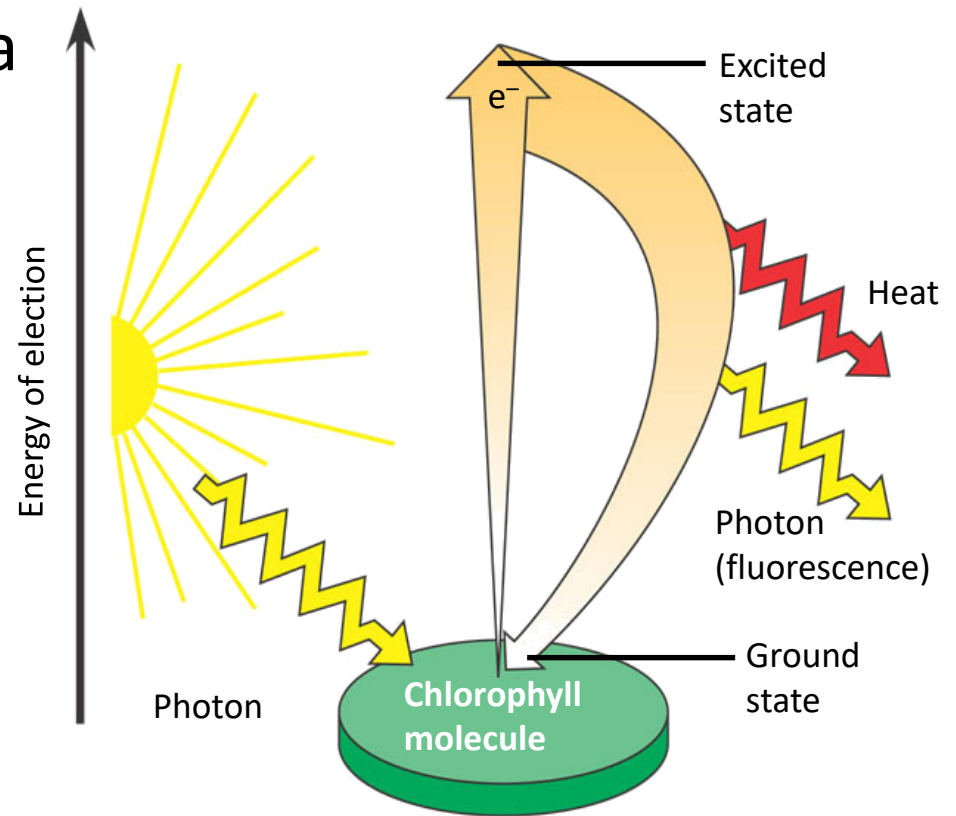
Photosynthetic pigments:

Extra Info

- Primary pigments: P680 & P700
 - Chlorophyll *a* that absorb light with wavelength 680 nm & 700 nm
 - Accessory pigments: chlorophyll *a* & *b* other than P680 & P700

Light excites chlorophyll

- Each chlorophyll molecule can absorb a photon of light, get excited & energized



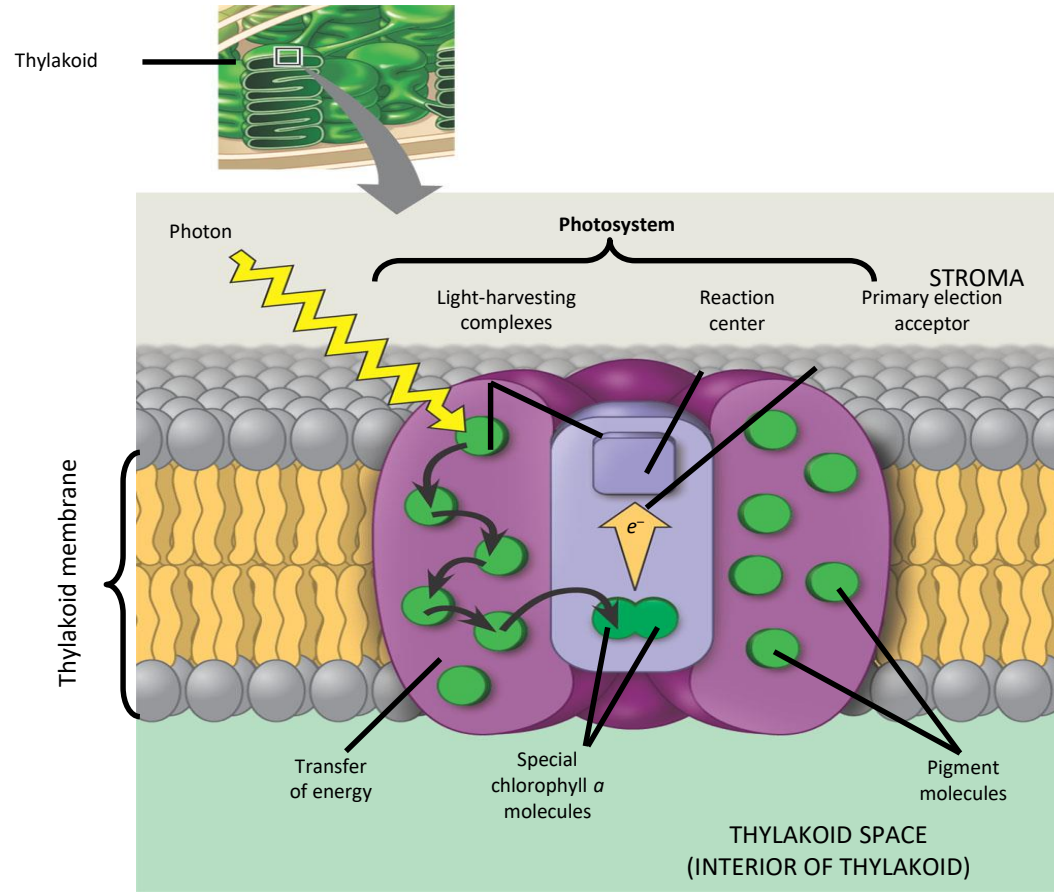
Photosystem

- Pigments are organised into photosystems
- Cluster of photosynthetic pigments grouped together in the form of particles/ lipid protein globules in the internal membrane of thylakoid & intergranal lamellae inside chloroplast
- Each have 200 – 300 molecules of chlorophyll, accessory pigments & proteins within a protein matrix – antenna for gathering light

Photosystem

- 2 types
 - I. Photosystem I
 - II. Photosystem II

Photosystem



Photosystem I

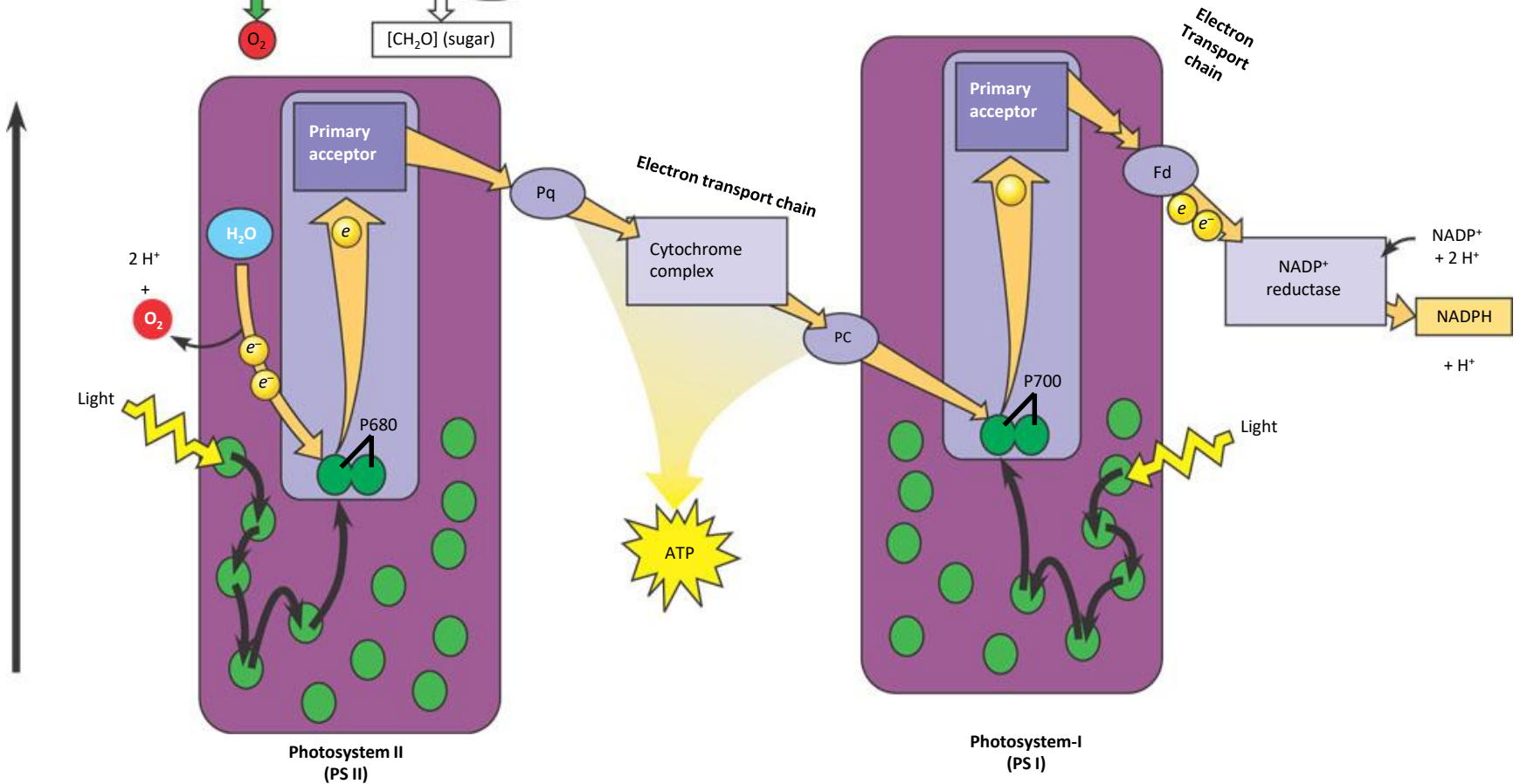
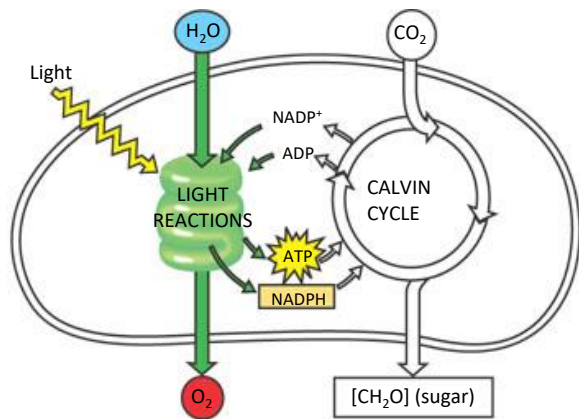
- A system of pigments in smaller particles
- Found largely in intergranal lamella
- 1st photosystem evolved
- λ of light most strongly absorbed is 700 nm
- Has 1° chlorophyll a, P700 in its reaction centre

Photosystem II

- A cluster of pigments in the form of bigger particles
- Found in the thylakoid membrane of grana
- Later photosystem evolved
- λ of light most strongly absorbed is 680 nm
- Has 1^o chlorophyll a, P₆₈₀ in its rxn centre

Light Dependent Reactions

- Light is absorbed by PS II & PS I in thylakoid membranes
- e⁻ flows via ETC in 2 ways:
 - I. Noncyclic photophosphorylation
 - II. Cyclic photophosphorylation



Noncyclic Photophosphorylation

- e- enter 2 ETC
- ATP & NADPH (nicotinamide dinucleotide phosphate) are formed

Photoactivation of PS II

- Pigments (accessory) in PS II absorb light, get excited
- E is raised, passes from one to another until rxn centre, e⁻ is released
- e⁻ is transferred to plastoquinone – PQ (e⁻ acceptor)
- PQ- a complex wth 2 cytochromes & othr proteins
- To a system of e⁻ carriers (b6-f complex) – as proton pump

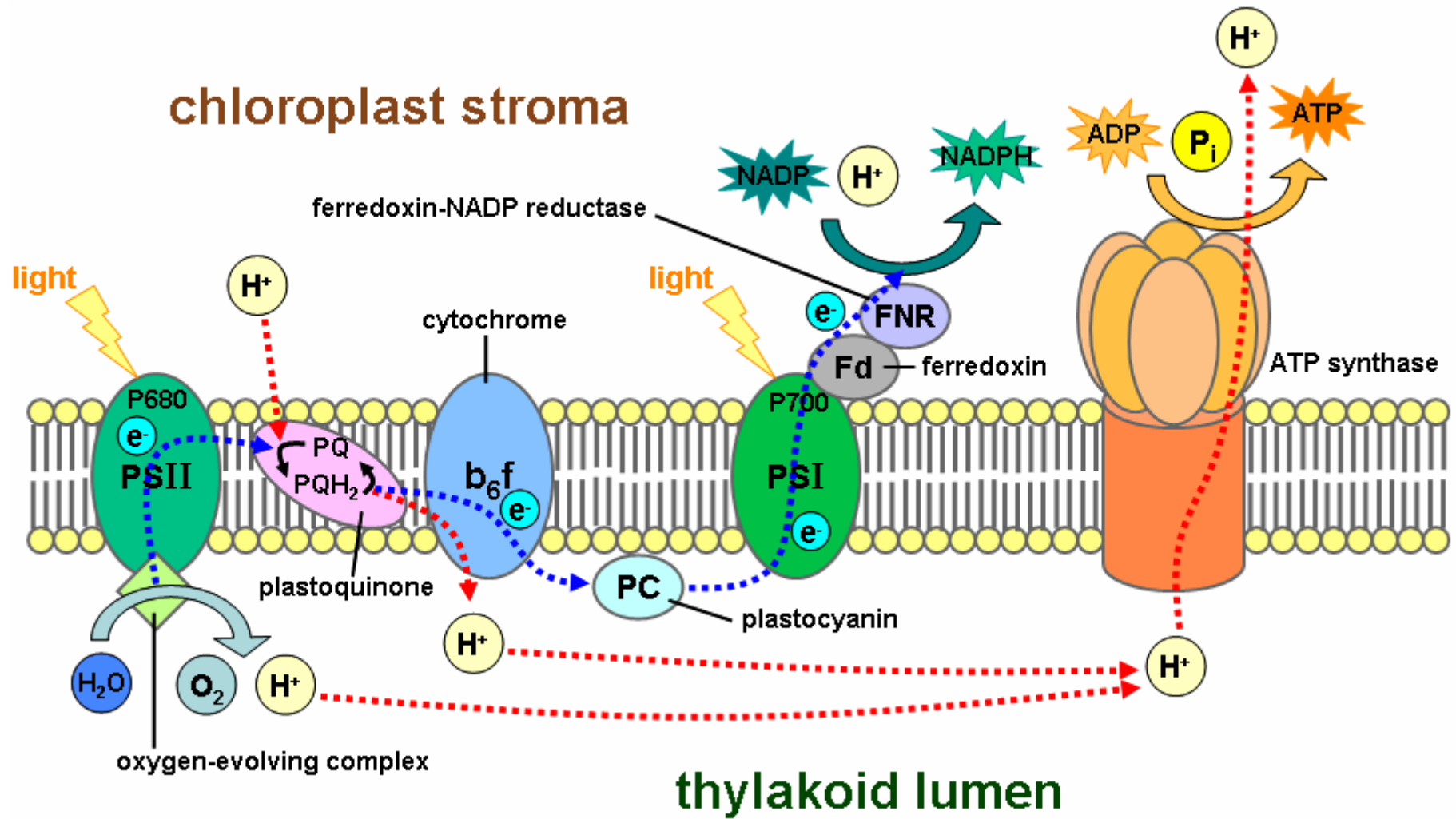
- Proton pump, pumps proton into thylakoid space (lumen)
- Proton are then used to form ATP from ADP & Pi (Photophosphorylation)
- e- back to PS II from photolysis of H₂O
- $2 \text{H}_2\text{O} \rightarrow 2 \text{H}^+ + \text{O}_2$
- H ions increase the proton gradient inside thylakoid

- Flows of e^- is exergonic & provide E to produce ATP by chemiosmosis
- ATP is formed as proton diffuse down [] from thylakoid space via ATP-synthase channels into stroma
- ATP produced provides E to Calvin cycle
- NADP reduced to formed NADPH by picking up 2 protons released in photolysis PS II
- NADPH carries H to Calvin Cycle to make sugar

Photosystem I – P₇₀₀

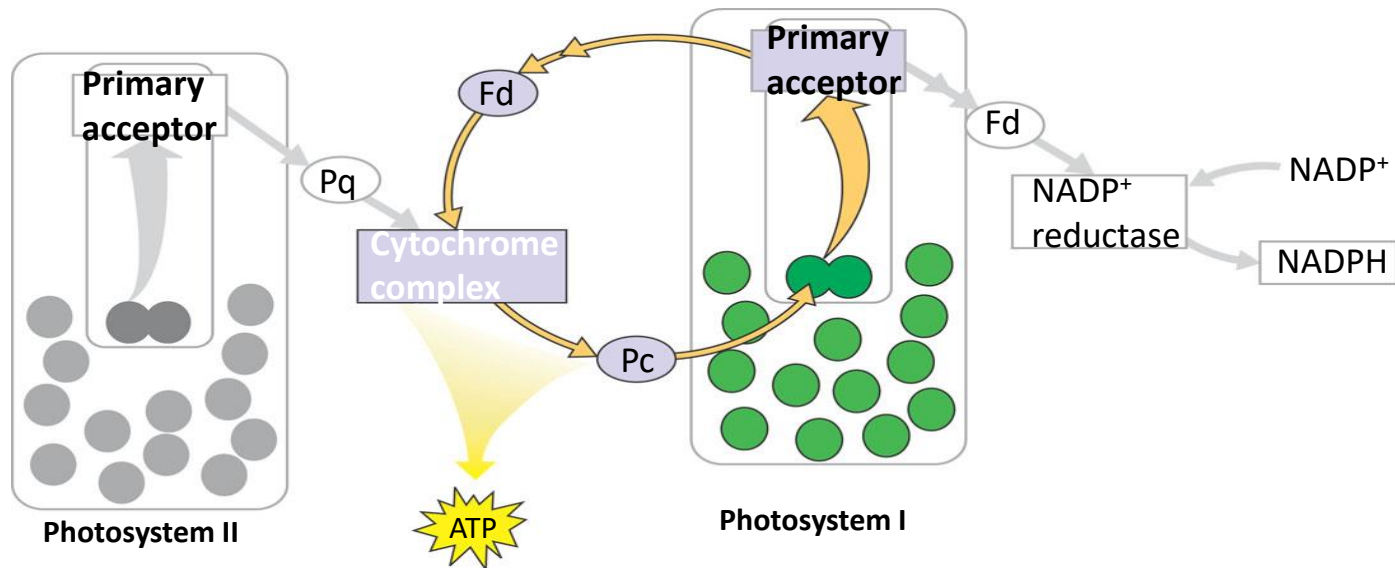
- E is absorbed by P700
- e⁻ from the head of chlorophyll a become excited & energized, captured by primary e⁻ receptor
- e⁻ is replaced by PS II
- ETC has ferredoxin (FD)
- End product is NADPH, no ATP

chloroplast stroma



Cyclic Photophosphorylation

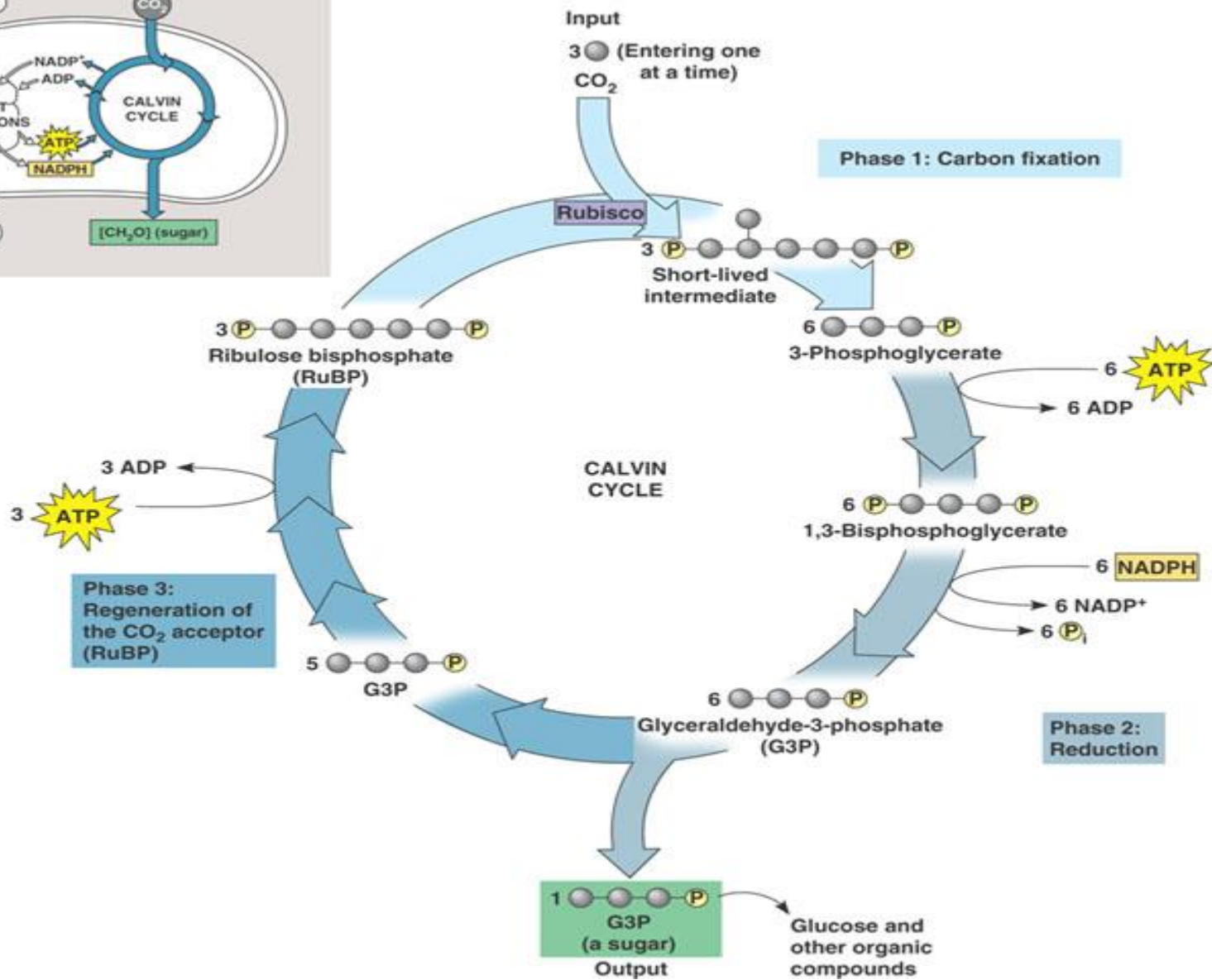
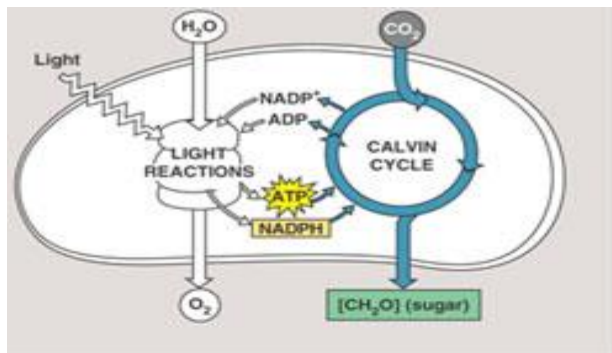
- In cyclic electron flow
 - Only photosystem I is used
 - Only ATP is produced



- Production of sugar in Calvin cycle uses a lots of ATP
- When chloroplast low In ATP, cyclic photophosphorylation occurs to replenish ATP levels
- e- frm PS II to PS I, primary e- acceptor, cytochrome complex PS II

Calvin cycle

- The Calvin cycle
 - Is similar to the citric acid cycle
 - Occurs in the stroma
 - Cyclical process produces 3-carbon sugar phosphoglyceraldehyde (PGAL)
- The Calvin cycle has three phases
 - Carbon fixation
 - Reduction
 - Regeneration of the CO₂ acceptor



Process of Calvin Cycle

- C enters the stomates in the form of CO₂ & becomes fixed (Carbon fixation)
- Is a reduction rxn, C is gaining H
- CO₂ enters Calvin Cycle, attached to a 5-carbon sugar, ribulose biphosphate (RuBP) forming 6-C molecules
- 6 – C molecules unstable : 2 3 –C molecules, 3-phosphoglycerate (3-PGA)
- Enzyme: Ribulose biphosphate carboxylase (rubisco)

Photorespiration

- On hot, dry days, plants close their stomata
 - Conserving water but limiting access to CO_2
 - Causing oxygen to build up
- In photorespiration
 - O_2 substitutes for CO_2 in the active site of the enzyme rubisco
 - The photosynthetic rate is reduced
 - No ATP produced

C - 3 Plants

- Plants that CO₂ enters Calvin cycle & is fixed into 3-phosphoglycerate (3-PGA) by Rubisco

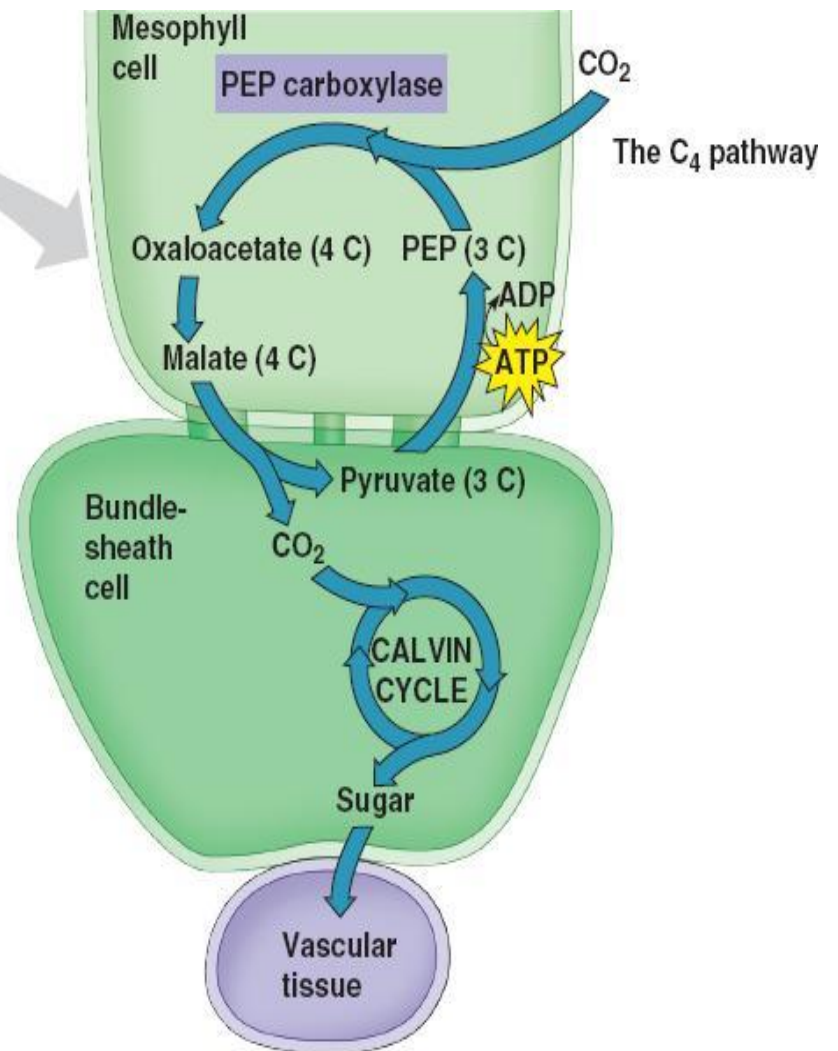
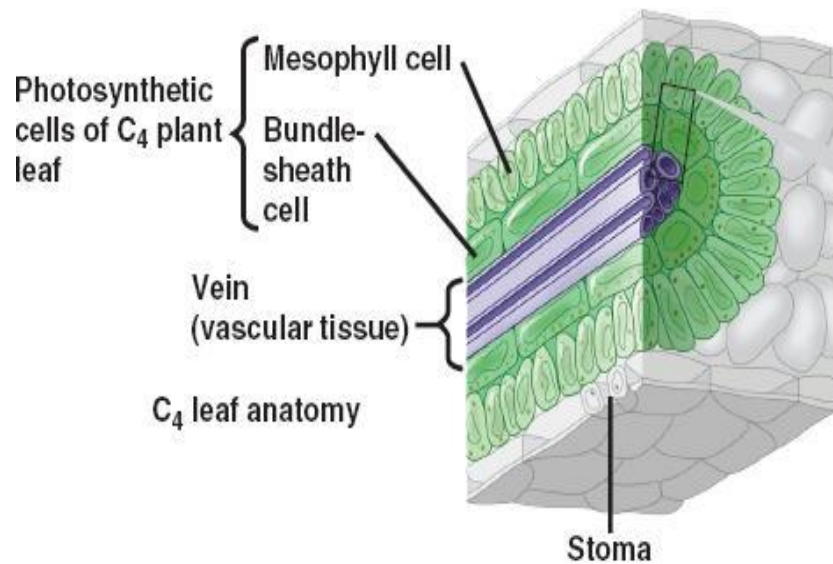
C₄ Plants

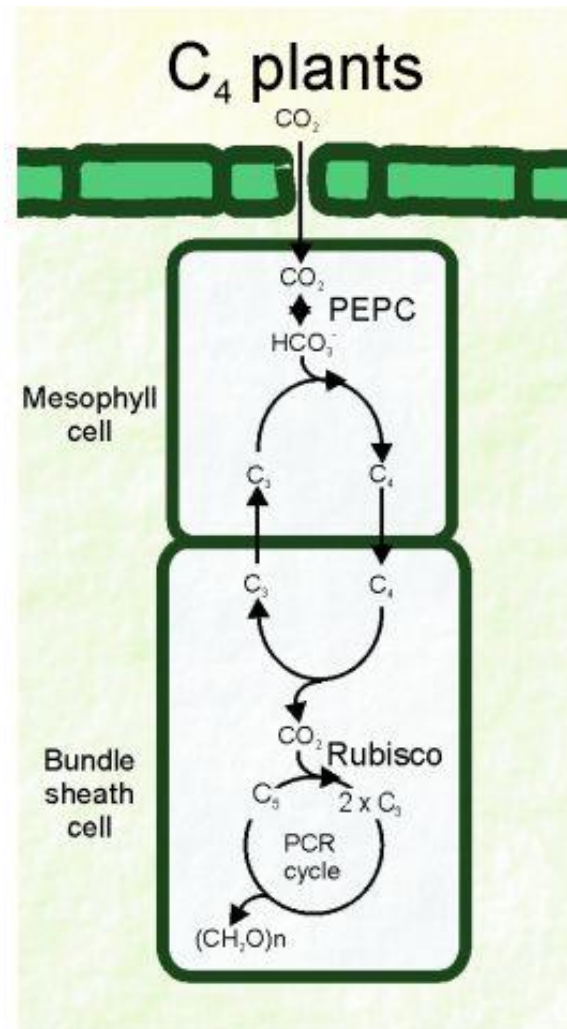
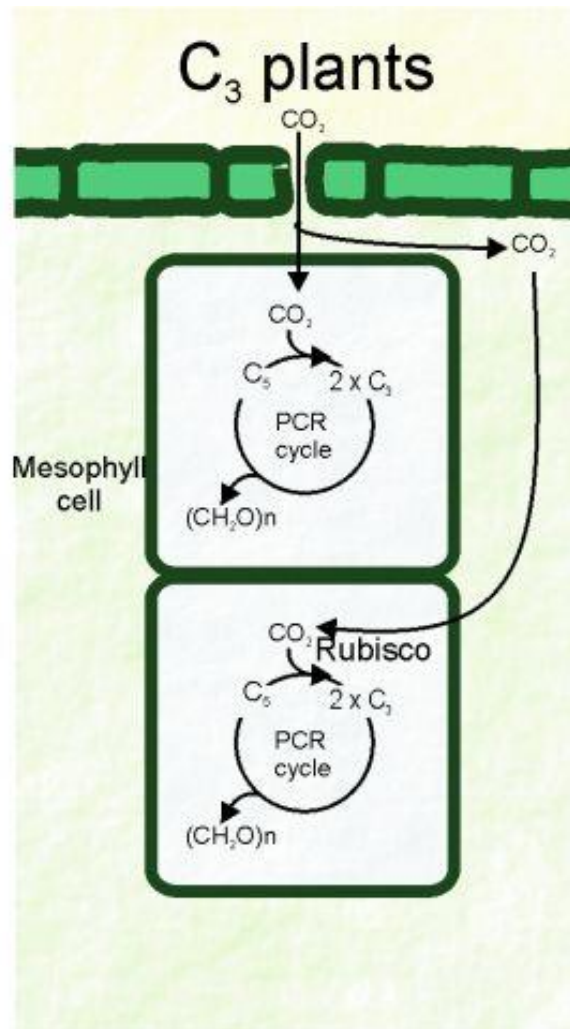
- C₄ plants minimize the cost of photorespiration
 - By incorporating CO₂ into four carbon compounds in mesophyll cells
- These four carbon compounds
 - Are exported to bundle sheath cells, where they release CO₂ used in the Calvin cycle
- Adapt to hot & sunny environment
- Corn, sugar cane, crabgrass

C₄ Pathway

- CO₂ enters mesophyll, combines with a 3-C molecule, phosphoenolpyruvate (PEP) to form 4 –C molecule oxaloacetate by PEP carboxylase
- Then form malic acid/malate , it pumps via plasmodesmata into adjacent bundle sheath cell
- Release CO₂, then enter Calvin Cycle, combine th PGAL
- Rubisco can fix CO₂ better , less photorespiration
- Hatch- Slack pathway

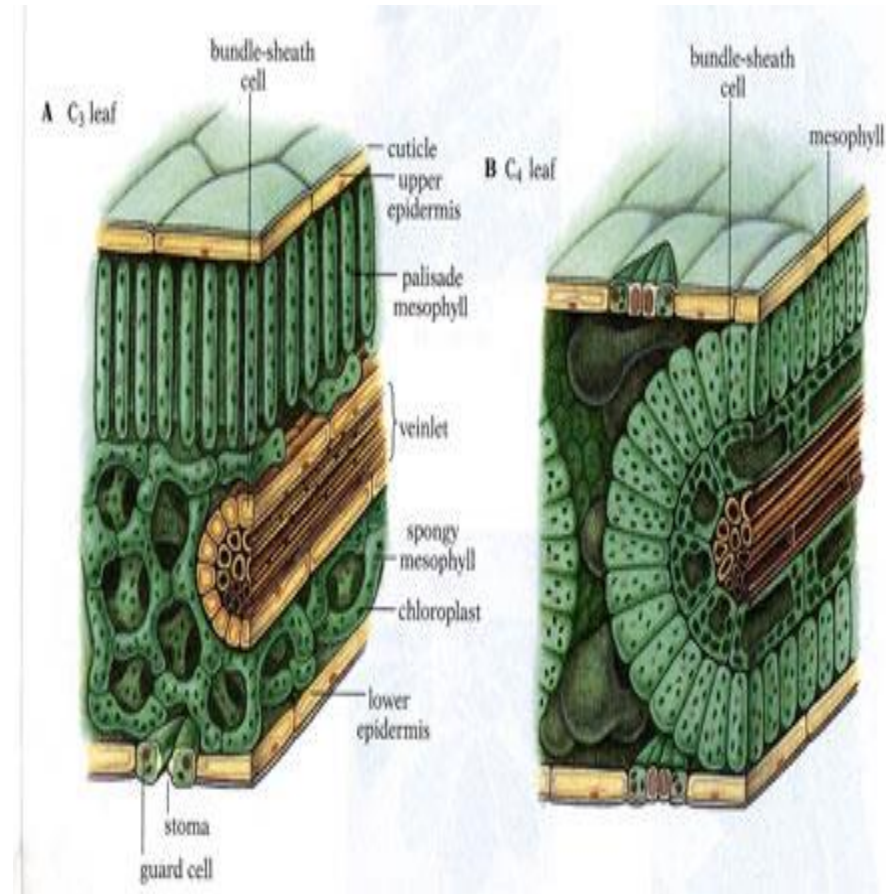
- C-4 plants can maximize the amount of CO₂ that diffuse into air space
- Minimize the length of stomates must remain open





Kranz Anatomy

- C- 4 bundle sheath cells lie under mesophyll cells
- Light rxn in mesophyll
- Dark rxn in bundle sheath cells

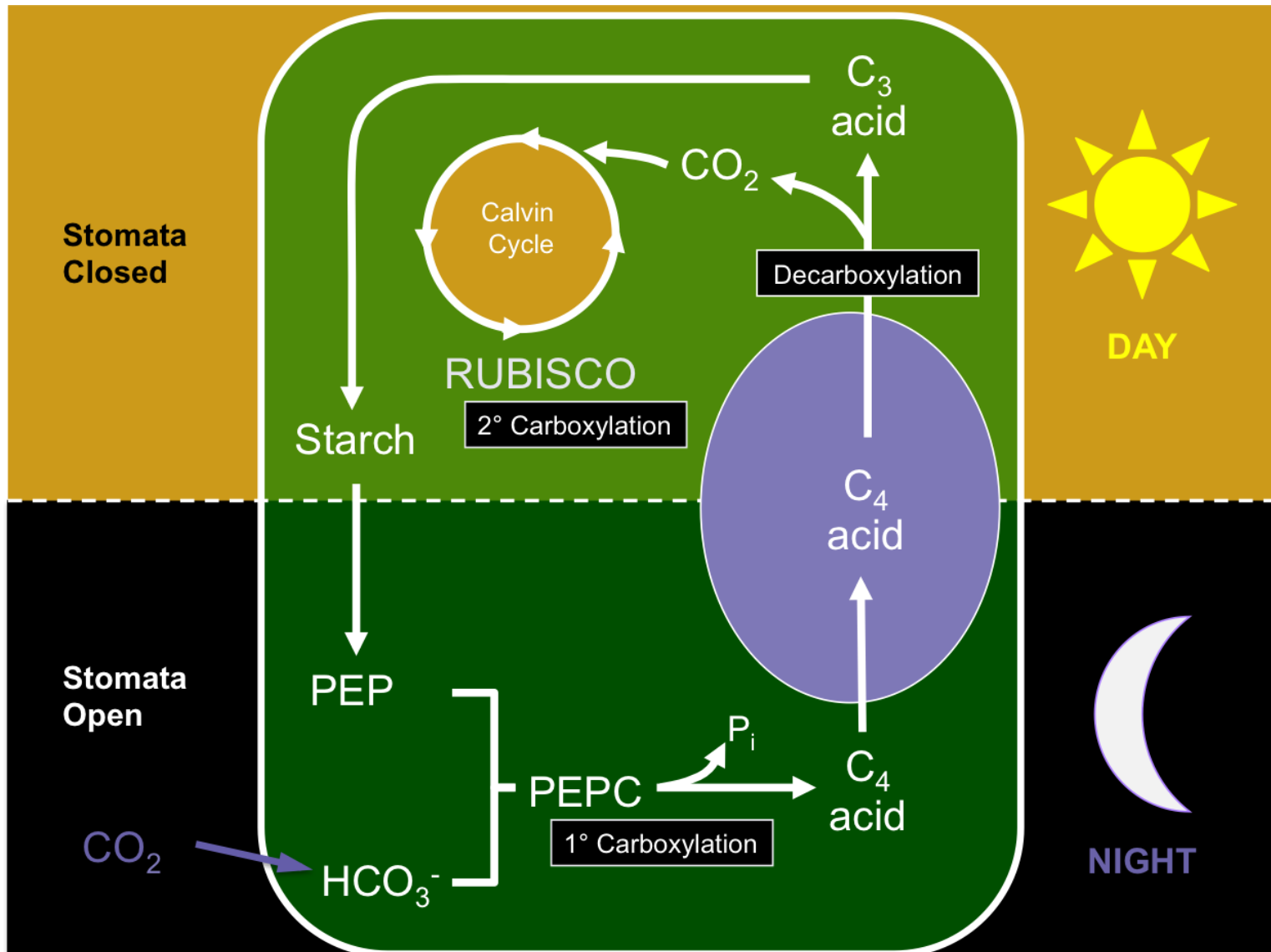


CAM Plants

- Belong to family of Crassulaceae includes *Bryophyllum*, open stomata drg night, close drg day
- Pineapple, cacti
- Crassulacean acid metabolism (CAM) occurs in mesophyll cells

- At night, stomata open, CO₂ is accepted by PEP to form oxaloacetate in Hatch-Slack pathway
- Oxaloacetate is reduced to malate requires NADPH
- Malate is stored in sap vacuole (night only)
- At day, malate release CO₂ by decarboxylation during photosynthesis

- CO₂ is fixed by RuBP to form PGA, undergoes Calvin Cycle, to form glucose & starch
- Pyruvate from decarboxylation of malate can be converted to starch & stored during day
- At night, starch is hydrolysed to maltose then glucose, converted into PEP & cycle starts again
- Enable plant to carry out photosynthesis despite stomata are closed during the day.



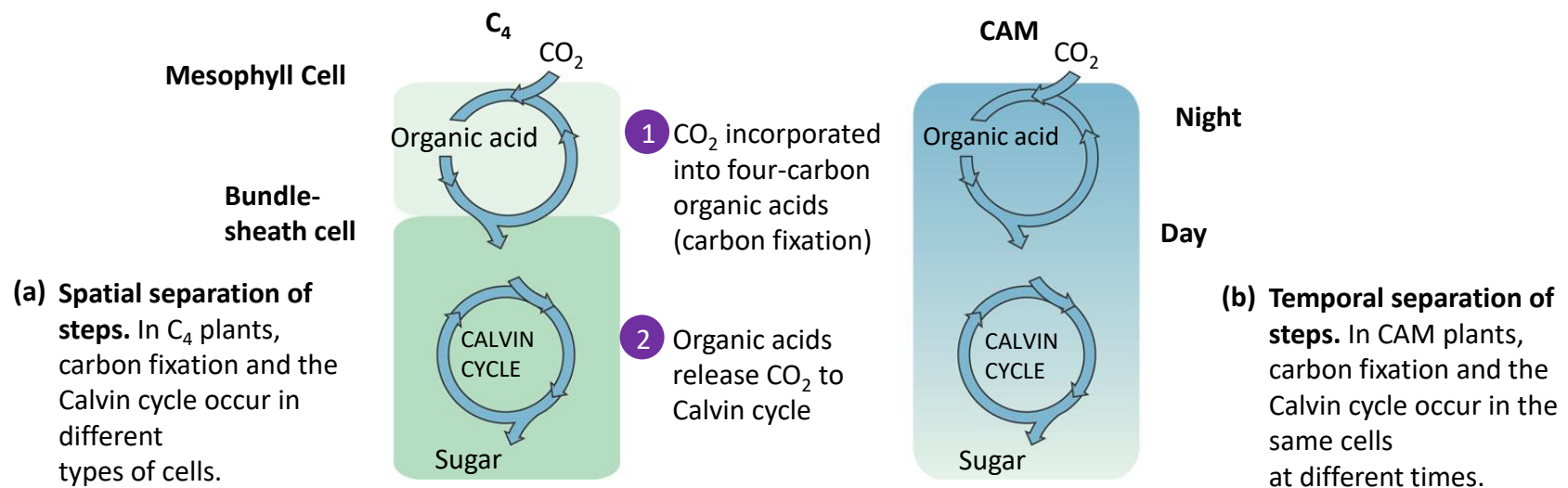
- The CAM pathway is similar to the C_4 pathway

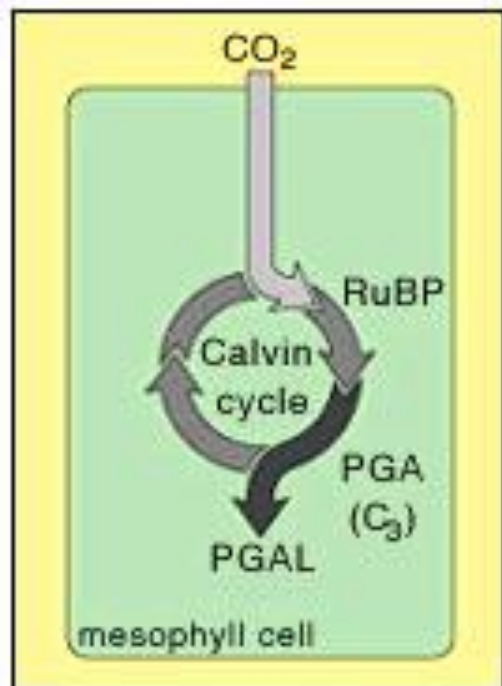


Sugarcane

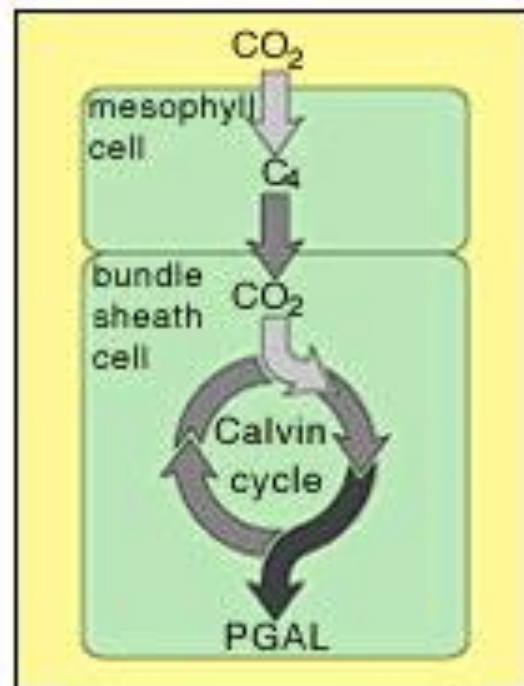


Pineapple

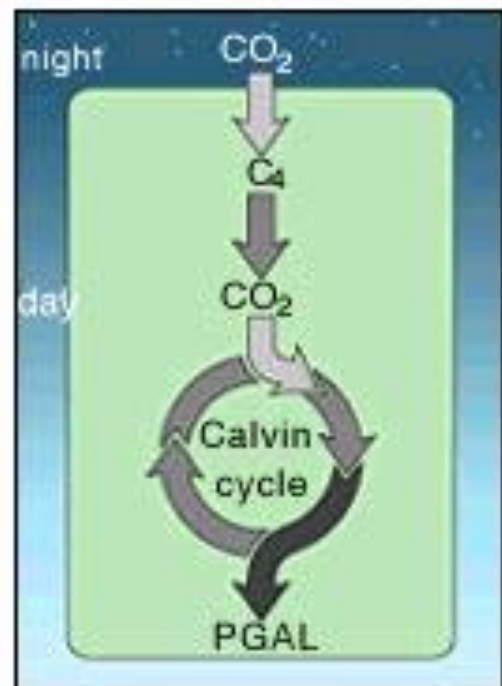




CO₂ fixation in a C₃ plant



CO₂ fixation in a C₄ plant



CO₂ fixation in a CAM plant

Thanks